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Purpose/Policy:

“This “Can Line Depalletizer” standard operating procedure has been created to assist operators with setup, operations, changeover, cleaning, and troubleshooting.

Scope:

The “Depalletizer” requires input from two operators. The downstairs of the “Depalletizer” is controlled by the Forklift operator upon start up. The E-Stop next to the can load bay is pulled and a pallet of cans is loaded onto the line. The upstairs operator is responsible for the formal operation of the unit.

The purpose of the “Depalletizer” is to de-pallet raw can material in preparation for an upcoming can run. The “Depalletizer” brings in a pallet of cans and then uses a lift to bring cans up to the second floor of the can line. Once the pallet ring has been removed from the pallet, the machine can remove slip sheets and push rows of cans onto the conveyor belt system.

Definitions/Acronyms:

- SOP – Standard Operating Procedure
- PPE – Personal Protective Equipment
- HMI – Human Machine Interface (the touch screen used to control a machine)
- E-Stop – Emergency Stop button
- Depalletizer (Depal) – Machine that raises pallets of empty cans to the second floor and pushes rows of cans onto the can line conveyor

Authorities/Responsibilities:

“This SOP can be completed by an entry level packaging operator after proper training from a senior member of the Can line packaging team. The objective of this SOP is to gain the knowledge needed to operate the “Depalletizer” and trouble shoot any issues that may arise.”

Resources and Materials:

Operations:

- Can Guide.

Changeovers:

- Twist Knob for height change– attached to the palletizer.

Basic Cleaning:

- Dry rags or paper towel.
- Soapy Water.

Deep Clean:

- Refer to SOP Document - Can line Zone 1 Cleaning
  - N:\Operations\Production\Boston\Packaging\Canning Line\Canning 2025\Zone Cleaning 2025\Can Line Upper Level\Zone 1 – Depal

Troubleshooting:

- No Equipment required.

PPE Required:

- No formal PPE required suggested nitrate gloves for cleaning.

Procedures:

Lower-Level Operations – Pallet Loading:

1. A forklift certified lift driver will load a pallet of raw cans for whatever run is needed onto the Depal infeed conveyor.
2. Cut the binding straps.
3. Ensure the Filler/Lift operator has the Depal ready to start.
4. Press the full pallet release button located on the side panel of the machine.
   4.1 *NOTE*: Full pallet release can also be pressed upstairs on the HMI.

Upper-Level Operations – Production Run:

1. From the HMI upstairs, ensure the line is clear and undo the E-Stop
2. Press “control power” to boot up the HMI.
3. Ensure that the “Depalletizer” is set to the correct size can.
   3.1 Please refer to the “Changeover Guide” section of this SOP for changeover instructions.
4. If the Depal is in the correct size, from the HMI tap the “MANUAL MODE” button to toggle it into “AUTO MODE”.
   4.1 *NOTE***:** Auto and manual are the same button, tapping it will toggle between the two modes of operation.
5. Press the Green “Cycle Start” button on the HMI.
6. Once the pallet has been brought up to the top and the depal stops, remove the top ring from the raw pallet.

6.1 *NOTE*: If the can pallet is not loading into the “Depalletizer”, select “Full Pallet Release” from the HMI screen.

6.2 This will feed in and then bring up the staged pallet located on the infeed conveyor.

7. Toggle the” Accumulation Table” and “Grippers” into auto by selecting them on the HMI screen.
8. Place the “Can Guide”, located under the HMI, on the stainless-steel edge of the accumulation table, adjacent to the pallet. This will provide pressure and prevent cans from tipping as they leave the pallet and are transferred onto the table.
9. Press “Start” on the HMI

9.1 This will bring up a layer of cans and push them onto the accumulation table. You can now remove the can guide and return it to its shelf.

10. After this process is complete, the machine will run automatic unless otherwise changed to manual mode.

Changeover Guide:

Changing over the “Depalletizer” between 12oz, 16oz and 19.2oz cans requires you to move a photo-eye. The eye is located at the middle of the depal holding position upstairs and has a guide marker for each size.

1. Loosen the orange “Twist Knob” and move the photo-eye down the marker to the stop for 12oz.
2. Loosen the “Twist Knob” and move the photo-eye up halfway up the marker stop and insert the pin to select 16oz. Loosen the pin by engaging the push button.
3. Loosen the “Twist Knob” and move the photo-eye all the way up the marker for 19.2oz.

Ending the Production Run (Runout):

1. When nearing the end of a run, on the HMI select “End of Run” – commonly referred to as run-out mode. This will slow the swipe of cans to keep the accumulation table mostly empty, preventing overpopulation of the line and having to recycle excess cans.
2. Once the run is complete, toggle the machine mode into manual by tapping on the “AUTO MODE” button on the HMI.

2.1 *NOTE:* Auto and manual are the same button, tapping it will toggle between the two modes of operation.

3. Ensure the pusher is in the forward position.
4. Select manual screen from the HMI.
5. Place a Slip-Sheet on top of the cans still on the pallet, followed by a Pallet-Ring.
6. Using manual mode, lower the pallet down by selecting and holding “Hoist Down” button.

7. Once the pallet has made it to the bottom, press and hold the “Infeed Pallet Conv Rev” button on the HMI until the pallet is clear of the hoist and out far enough for the lift driver to grab.
    7.1 Be sure to jog the pallet clear of the hoist safety photo eyes, stopping while they are blocked will result in an inability to eject a pallet using the conveyor.
        7.1.1 *NOTE:* Be sure not to send the pallet past the edge of the conveyor, as it may fall.
8. Once the pallet has been ejected all the way, press the E-Stop on the control panel.

Basic Cleaning of Depalletizer:

1. Wipe down all areas except for the HMI with a lightly damp rag.
2. Dry area using a soft dry cloth.
    2.1 *NOTE:* Stainless Steel cleaner / Alpha 5 can be used to help with heavier soiled areas.

Deep Cleaning Guide for Depalletizer:

1. Link to Refer to SOP Document - Can line Zone 1 Cleaning
    1.1 N:\Operations\Production\Boston\Packaging\Canning Line\Canning 2025\Zone Cleaning 2025\Can Line Upper Level\Zone 1 – Depal.

Forms:

Not Applicable

Reference Documents:

Not Applicable

Common Issues:

Pallet Jam:

A pallet jam can occur when the can pallet infeed’s too far into the hoist area or when a pallet does not eject far enough from the hoist area.

1. Check the hoist area on the ground floor to ensure no eye is blocked and the pallet has not over traveled.
2. After ensuring neither of these conditions are met and that all eyes are clear, you should be able to restart the depal from the HMI.
    2.1 *NOTE:* Sometimes the sensor is knocked out of position, move it back so that it’s looking at the pallet discharge.

Machine not running:

1. Ensure the machine is in auto mode including grippers and accumulation table. When one of these or both are off, the Depal will not run in Auto.

Hoist doesn't bring up pallet (no fault on HMI):

1. Check gripper sensor located under the slip sheet holding table.
2. If the pallet is not coming into the lift area check to make sure the elevator is all the way down.
3. The photo-eye should light up as pusher retracts.
4. Make sure the pusher is in the forward position.
    4.1 If this condition is met, the machine may require a power cycle.
        4.1.1 *NOTE*: Do not power-cycle the machine, contact maintenance to do-so.

Document History and Approval:

Revise document history as described in the table and key below.

(Reference to following document: B1C.SOP.P001: Document Control and Approval Procedure Template)

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Approved by: Kyle Szymanski

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